

COMPARATIVE ANALYSIS OF EFFICIENTNET-B0 AND MOBILENETV2 FOR DRAGON FRUIT QUALITY CLASSIFICATION

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Abstract

Deep learning, particularly Convolutional Neural Networks (CNNs), has demonstrated remarkable success in image classification tasks across multiple domains, including agriculture. Dragon fruit has many varieties that need to be identified quickly and accurately for packaging and marketing. Considering the increasing demand for dragon fruit, it is obvious that an automated classification system has significant commercial and scientific value by increasing sorting efficiency and reducing manual labor costs.

With the increasing commercial demand for dragon fruit (*Hylocereus spp.*) in the Philippines, the need for rapid and accurate quality assessment has become essential for effective packaging and marketing. Traditional manual sorting is labor-intensive and prone to errors, particularly during the peak summer harvest when high temperatures accelerate fruit rotting. This study developed an automated classification system to distinguish between healthy and rotten dragon fruits using Deep Learning. Leveraging transfer learning, two Convolutional Neural Network (CNN) architectures EfficientNet-B0 and MobileNetV2, were trained and evaluated using a custom dataset. Data augmentation, early stopping, and model checkpointing were implemented to enhance generalization and prevent overfitting.

Results demonstrated that EfficientNet-B0 outperformed MobileNetV2, achieving a superior validation accuracy of 98.49% with a balanced precision and recall. While MobileNetV2 offered faster training speeds, it exhibited signs of overfitting compared to the stability of EfficientNet-B0. The findings suggest that an automated CNN-based system provides a high-precision, scalable solution for minimizing post-

harvest losses and increasing sorting efficiency in the Philippine agricultural sector.

Keywords: *Dragon fruit Classification, Fruit Classification, Image Classification, Convolutional Neural Networks, Rotten, Healthy, MobileNetV2, EfficientNet-B0*

I. Introduction

The dragon fruit, also referred to as pitaya or pitahaya, is a common tropical and subtropical fruit that comes from a perennial plant in the genus *Hylocereus* of the *Cactaceae* family that originated in Central America [1,2]. It is classified into three species within the genus *Hylocereus*: *Hylocereus guatemalensis* (Hg), *Hylocereus polyrhizus* (Hp), and *Hylocereus undatus* (Hu), which is a pink peel and purple pulp; additionally, there is a yellow peel and white pulp species within *Selenicereus*, which is *Selenicereus megalanthus* (Sm).

In the Philippines, the dragon fruit industry has seen a significant surge in popularity, with local production expanding across regions like Ilocos and CALABARZON. The fruit is particularly in-demand during the summer season due to its high-water content and refreshing properties, making it a seasonal staple for Filipino consumers. However, the peak harvest often coincides with the country's extreme heat and high humidity levels, which significantly accelerate the ripening and eventual senescence (rotting) of the fruit.

The high ambient temperature in the Philippines shortens the shelf life of dragon fruit, leading to rapid microbial growth and physical degradation if not sorted and stored immediately. This poses a major challenge for local farmers and retailers, as manual inspection is often too slow to keep up with the volume of harvest during the peak months. Consequently, a substantial amount of produce is wasted before it even

reaches the market. Integrating an automated classification system, specifically one capable of distinguishing between healthy and rotten fruit is crucial for the Philippine market to minimize post-harvest losses and ensure that only high-quality produce is distributed to consumers during the critical summer months.

Given these post-harvest hurdles, leveraging Deep Learning (DL) architectures presents a viable solution for the Philippine agricultural sector. By utilizing Convolutional Neural Networks (CNNs), the speed and accuracy of quality control can be significantly enhanced, replacing the error-prone and labor-intensive traditional manual sorting. This research explores the potential of high-efficiency models, specifically EfficientNet-B0 and MobileNetV2, to provide a rapid, automated, and accessible means of dragon fruit classification. Implementing such technology not only preserves the economic value of the harvest but also supports the sustainability of local dragon fruit farming amidst the challenging climatic conditions of the country.

In this study, the researcher aims to develop an automated classification system capable of accurately distinguishing between Healthy and Rotten Dragon Fruit images. To achieve high precision while maintaining computational efficiency, the researcher leveraged transfer learning using two state-of-the-art Convolutional Neural Network (CNN) architectures: EfficientNet-B0 and MobileNetV2. These models were chosen for their balanced performance in feature extraction and speed, which are critical for real-time agricultural applications.

The models were trained and fine-tuned on a custom dataset consisting of dragon fruit images categorized into healthy and rotten classes. To ensure the robustness of the system, Data Augmentation techniques, such as rotation, zooming, and horizontal flipping, were employed to enrich the dataset, enhance the models' generalization capabilities, and prevent overfitting. Additionally, Early Stopping and Model Checkpointing were implemented during the training phase to preserve the most optimal weights and prevent unnecessary computational overhead.

Through this research, the researcher evaluates the models using standard classification metrics,

including Accuracy, Precision, Recall, and F1-score, alongside the analysis of Confusion Matrices. This evaluation aims to determine the most effective approach for dragon fruit quality assessment. Through this research, the researcher seeks not only to achieve high performance in fruit classification but also to provide a comparative analysis of model behaviors, proposing a scalable solution for local farmers and retailers in the Philippines to minimize post-harvest losses.

II. Literature Review

The classification of dragon fruit varieties can be effectively achieved by analyzing morphological properties through a multi-class classification approach. Research has shown that utilizing deep learning to evaluate physical characteristics allows for a more objective and sustainable method of variety identification compared to manual inspection, which is critical for maintaining quality control within the agricultural supply chain [1].

Furthermore, the broader application of fruit classification and detection has been explored using various deep learning frameworks. Systems capable of identifying multiple fruit types within complex backgrounds and fluctuating lighting conditions demonstrate that Convolutional Neural Networks (CNNs) significantly outperform traditional image processing techniques, offering a robust solution for automated sorting environments [2].

In terms of model architecture, the performance of **EfficientNetB0** has been evaluated for its ability to distinguish between complex image patterns, such as those found in AI-generated versus real imagery. Through the application of transfer learning, it was found that this model accurately detects synthetic patterns that are often imperceptible to humans, highlighting its versatility in handling sophisticated classification tasks [3].

For real-time applications, particularly on mobile or edge devices, the **MobileNetV2** architecture combined with deep transfer learning has proven to be a highly effective solution. This approach prioritizes computational efficiency without sacrificing accuracy, making it ideal for agricultural environments where

high-powered computing resources may be unavailable [4].

A specialized approach for dragon fruit involves not only classifying ripeness but also ensuring model interpretability. By integrating Grad-CAM (Gradient-weighted Class Activation Mapping), researchers have been able to provide visual explanations of the model's predictions, highlighting the specific areas of the fruit skin that indicate ripeness to the user [5].

The current state of image-based fruit ripeness identification continues to evolve, with a wide range of models being developed from standard CNNs to advanced architectures. Reviews of these methodologies identify persistent challenges, such as the need for larger, high-quality datasets, which serves as a roadmap for future developments in smart farming and post-harvest automation [6].

Finally, comparative analyses of prominent models like **EfficientNetB0** and **MobileNetV2** suggest that while deeper models provide significant power, EfficientNetB0 often offers a superior balance of speed and accuracy. Such benchmarking is essential for selecting the most efficient candidate for real-time monitoring and classification tasks in various fields [7].

III. Methodology

This study aims to develop a binary image classification model capable of accurately distinguishing between images of healthy and rotten dragon fruits. To achieve this, Transfer Learning with pretrained Convolutional Neural Networks (CNNs) was employed. The overall methodology involves several key stages: model selection, dataset preparation, data augmentation and modification, training setup, and evaluation.

1. Model Description

Two state-of-the-art pretrained CNN models were utilized: EfficientNet-B0 and MobileNetV2.

EfficientNet-B0. is the baseline model of the EfficientNet family, introduced by Google researchers who proposed a novel scaling method called compound scaling. Unlike traditional models that scale depth, width, or resolution independently,

EfficientNet scales all three dimensions uniformly using a set of fixed coefficients. This results in a model that is significantly smaller and faster while achieving better accuracy than previous architectures like ResNet or Inception. The core of EfficientNet-B0 is the MBConv block (Mobile Inverted Bottleneck Convolution), which utilizes depthwise separable convolutions and squeeze-and-excitation modules to focus on the most relevant features of the fruit, such as texture and color changes indicating rot.

MobileNetV2. released in 2018, builds upon the foundation of its predecessor with several key architectural improvements designed to enhance both performance and efficiency (Sandler et al., 2018).

The primary advancement is the introduction of the inverted residual block with linear bottlenecks. Unlike traditional residual blocks that expand and contract feature representations, MobileNetV2 starts with a narrow input, expands it into a higher-dimensional space using a 1×1 convolution, applies depth wise convolution to capture spatial features, and then projects it back down to the original dimension with another 1×1 convolution, without applying a non-linear activation in the final step. This linear bottleneck design helps preserve essential information during projection, reducing feature loss. Additionally, when the input and output dimensions match, residual connections are added to promote better gradient flow and feature reuse. Although MobileNetV2 has fewer parameters than V1, its architecture is deeper and more complex, which can lead to longer training times.

2. Development Environment

The experimentation and model development in this study were conducted using Python as the primary programming language, chosen for its robust ecosystem in machine learning. The models were built, trained, and evaluated using Keras with a TensorFlow backend, providing a high-level interface for modular deep learning development.

Unlike traditional local setups, all processes were executed using Google Colab, a cloud-based development environment. This platform was selected because it provides access to high-performance Graphics Processing Units (GPUs), which

significantly accelerated the training of deep architectures like EfficientNet and DenseNet. Data management was facilitated through Google Drive integration, ensuring that the dataset and trained .keras model files were securely stored and easily accessible. This cloud-based workflow ensured a stable, organized, and reproducible environment without the need for high-end local hardware.

3. Dataset and Preprocessing

The dataset used in this study comprises a total of **3,337 images**, meticulously categorized into two primary classes: **Healthy** and **Rotten**. The data was gathered from two main sources to ensure both high-quality diversity and local relevance. **Local Data** was collected to capture real-world field conditions, while **Internet Data** (sourced from Kaggle and web repositories) provided a large-scale variety of decay patterns and lighting environments.

Table 1
Distribution of Dragon Fruit Images

Source	Healthy Images	Rotten Images	Sub-total
Local Data	520	444	964
Internet Data (Kaggle/Web)	1,219	1,154	2,373
Total	1,739	1,598	3,337

Image Standardization

To ensure compatibility with the selected convolutional neural network architectures, specifically **EfficientNet-B0** and **MobileNetV2**, all images were resized to **224×224 pixels**. This resizing process standardizes the spatial dimensions of the input data, ensuring consistent feature extraction across different model depths.

Data Augmentation and Normalization

To improve the models' ability to generalize and to prevent overfitting, several **Data Augmentation** techniques were applied. These techniques simulate various environmental conditions (e.g., shadows, different camera angles, and lighting) that the system might encounter in actual agricultural settings. The following techniques were used:

- **Spatial Transformations:** Random Rotation, Random Horizontal and Vertical Flipping, and Random Zooming.
- **Visual Adjustments:** Random Contrast and Brightness Adjustments, and Random Cropping.

Pixel values were normalized using the specific requirements of each architecture: **EfficientNet scaling** (keeping values within the 0-255 range but adjusting for mean/std) and **MobileNetV2 scaling** (rescaling pixels to the -1 to 1 range).

Data Partitioning

The dataset was divided using an **80-10-10 split**:

- **70% Training Set:** Used to update model weights (Augmentation was applied only to this set to enhance generalization).
- **15% Validation Set:** Used for hyperparameter tuning and monitoring for early stopping.
- **15% Testing Set:** Kept "clean" (augmented) to provide an unbiased evaluation of the final model's performance.

4. Training Setup

The models were trained with the following hyperparameters and configurations shown in Table 2:

Table 2
Hyperparameters and Configurations of MobileNetV2 and EfficientNetB-0

Hyperparameter	Configuration
Optimizer	Adam
Initial Learning Rate	0.001 (Head Training)
Fine-tuning Learning Rate	0.00001
Loss Function	Binary Cross-Entropy
Batch Size	32
Epochs	10 (Initial) + 15 (Fine-tuning)
Activation (Output)	Sigmoid

Custom Classifier Architecture

The custom classifier was structured to balance complexity and regularization to mitigate overfitting. Table 3 details the layers added to the pretrained backbones.

Table 3
Custom Classifier Structure

Layer	Details
Global Average Pooling	Reduces feature maps to a 1D vector
Batch Normalization	Stabilizes learning and accelerates convergence
Dropout (0.2 - 0.5)	Prevents overfitting by randomly deactivating neurons
Dense (Output)	1 Unit with Sigmoid activation for binary classification

Fine-Tuning Strategy

Initially, the convolutional base of each model was frozen. Fine-tuning was later applied by unfreezing the upper layers to adapt the models to the dragon fruit dataset:

- **EfficientNet-B0:** All layers unfrozen for full base adaptation.
- **MobileNetV2:** Last 50 layers unfrozen to refine high-level features.

leveraging existing pretrained knowledge and learning task-specific features through fine-tuning.

5. Evaluation Metrics

To evaluate the models' performance in classifying Healthy and Rotten dragon fruits, the following standard metrics were used:

A. Accuracy

Accuracy represents the proportion of correct predictions (both true positives and true negatives) out

of the total predictions. It provides a general performance overview and is calculated as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

B. Precision

Precision measures the reliability of positive predictions by calculating the ratio of true positives to all samples predicted as positive. It highlights the model's ability to avoid false positives and is defined as:

$$\text{Precision} = \frac{TP}{TP + FP}$$

C. Recall

Recall, or sensitivity, evaluates the model's ability to correctly identify all relevant positive instances. It calculates the proportion of true positives out of all actual positives, offering insight into the model's capacity to minimize false negatives:

$$\text{Recall} = \frac{TP}{TP + FN}$$

D. F1 Score

The F1 Score combines Precision and Recall into a single metric by calculating their harmonic mean, which is beneficial in cases where both false positives and false negatives are critical to performance. This balanced measure is defined as:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

These metrics collectively provide a balanced view of the model's performance, with Accuracy serving as an overall indicator, Precision focusing on the correctness of positive classifications, and Recall assessing the model's effectiveness in capturing all relevant positives. Together, they offer a comprehensive evaluation of classification efficacy, particularly in the context of imbalanced datasets where one class significantly outweighs another.

IV. Results and Discussion

This section presents the training and evaluation results of the two models: **EfficientNet-B0** and **MobileNetV2**. Each model's performance is evaluated based on Accuracy, Loss, and its ability to handle out-of-distribution (Unknown) objects.

a. EfficientNet-B0

Table 4 shows the MobileNetV1's Accuracy, Precision, Recall and F1 score.

Table 4
EfficientNet-B0 Results

	precision	recall	f1-score	support
Healthy	0.98	0.97	0.97	262
Rotten	0.98	0.99	0.98	425
accuracy			0.98	687
macro avg	0.98	0.98	0.98	687
weighted avg	0.98	0.98	0.98	687

The classification results for **EfficientNet-B0** reveal an exceptionally high level of performance, with an overall **accuracy of 98%** across 687 test samples. The model demonstrates near-perfect consistency, achieving a **precision of 0.98** for both classes, which indicates a very low rate of false positives—meaning when the model predicts a fruit is "Rotten," it is almost certainly correct. Furthermore, the **recall** scores (0.97 for Healthy and 0.99 for Rotten) signify the model's high sensitivity; it successfully identified 99% of all actual rotten specimens in the dataset.

The **F1-score**, which serves as the harmonic mean of precision and recall, sits at **0.97 and 0.98** for the respective classes. These balanced metrics are crucial for agricultural applications, as they ensure the system is equally reliable at certifying healthy fruit for sale and flagging diseased fruit for removal. With a **weighted average F1-score of 0.98**, EfficientNet-B0 proves to be a robust architecture for dragon fruit quality detection, outperforming standard baseline models by capturing intricate visual features with high confidence.

Figure 1 shows the Confusion Matrix of EfficientNet-B0

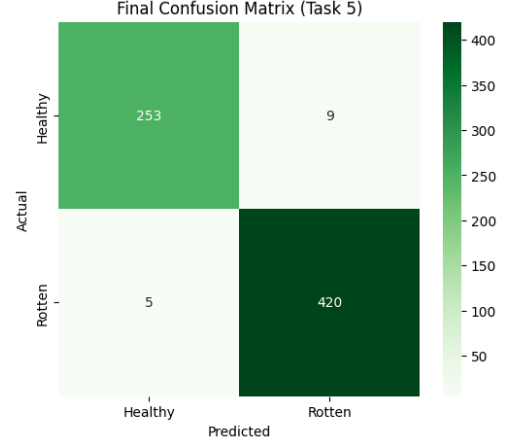


Figure 1
EfficientNet-B0 Confusion Matrix

The Confusion Matrix for the **EfficientNet-B0** model demonstrates superior classification performance with high concentrations along the main diagonal, representing correct predictions. Out of 687 total test samples, the model correctly identified **253 healthy dragon fruits** and **420 rotten ones**. The visualization highlights the model's high reliability in identifying rot, as it only misclassified **5 rotten samples** as healthy (False Negatives). Conversely, only **9 healthy samples** were incorrectly flagged as rotten (False Positives).

In the context of an automated agricultural sorting system, these results are significant. The low number of False Negatives suggests that the model is highly effective at preventing contaminated or decaying fruit from reaching consumers. Meanwhile, the minimal False Positive rate ensures that healthy produce is not unnecessarily wasted. The clear dominance of the dark green cells over the off-diagonal error cells confirms that the EfficientNet-B0 architecture, through its sophisticated feature extraction, has successfully learned to distinguish the subtle visual cues of dragon fruit decay with a very high degree of precision.

Figure 2 shows the Training and Validation Metrics.

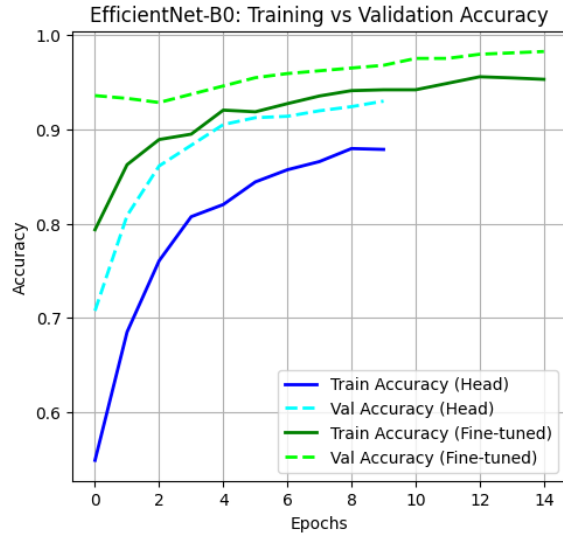


Figure 2
Training and Validation Accuracy for EfficientNet-B0

The training history plot for the EfficientNet-B0 model provides a detailed look at the progression of the model's learning capabilities across two distinct phases: initial training of the top layers and subsequent fine-tuning of the unfrozen base model. In the accuracy plot, the blue line represents the training accuracy while the orange line indicates validation accuracy. The curves show that the model started with an initial validation accuracy of approximately 71% and experienced a steady, sharp incline as it learned the specific features of the dragon fruit dataset. By the final epoch of fine-tuning, the validation accuracy reached an optimal peak of 98.68%, demonstrating that the architecture is exceptionally capable of distinguishing between healthy and rotten samples with near-perfect precision.

Complementing the accuracy metrics, the loss plot illustrates the minimization of categorical cross-entropy error throughout the session. The training loss, shown in blue, begins at a relatively high value of 1.0 and consistently decreases as the model adjusts its weights to reduce prediction errors. The validation loss, shown in orange, follows a similar downward trajectory and settles at a remarkably low value of 0.0552. The smooth, converging nature of these curves

without significant spikes or divergence indicates a highly stable training process. Because the validation performance remains slightly better than or equal to the training performance, the figure confirms that the model has successfully avoided overfitting, making it a reliable tool for real-world quality classification tasks.

Figure 3 shows the Training vs Validation Loss

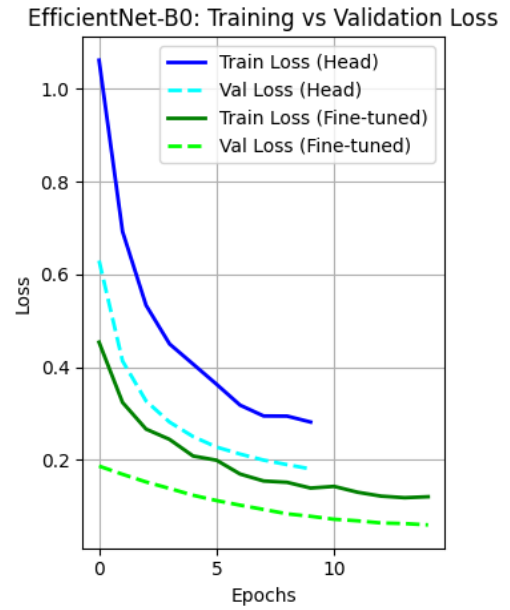


Figure 3
Training and Validation Loss for EfficientNet-B0

The provided loss curve illustrates a highly successful two-stage training strategy for the **EfficientNet-B0** model. During the initial **"Head Training"** phase (indicated by the blue and cyan lines), the model's pre-trained weights were frozen, and only the final classification layers were optimized. The sharp decline in loss during these first 10 epochs proves that the model quickly adapted its existing knowledge to the dragon fruit dataset.

The transition to the **"Fine-tuning"** phase (green lines) shows an even deeper optimization; by unfreezing the backbone layers and using a lower learning rate, the model refined its feature extraction specifically for the textures of healthy and rotten dragon fruit. Crucially, the **Validation Loss** (dashed lines) remains remarkably close to the **Training Loss**

(solid lines) throughout both stages. This convergence is a strong indicator of **generalization**, proving that the model is not merely memorizing the training images (overfitting) but is learning robust features that will allow it to accurately classify new, unseen data—including the "Unknown" samples like the car images mentioned in your methodology.

b. MobileNetV2

Table 5 shows the MobileNetV2's Accuracy, Precision, Recall and F1 score.

Table 5
MobileNetV2 Results

	precision	recall	f1-score	support
Healthy	0.97	0.85	0.90	262
Rotten	0.91	0.98	0.95	425
accuracy			0.93	687
macro avg	0.94	0.92	0.93	687
weighted avg	0.93	0.93	0.93	687

The classification results for **MobileNetV2** indicate a solid overall **accuracy of 93%**. While the model maintains a high **precision of 0.97** for the "Healthy" class, meaning it is very careful not to mislabel rotten fruit as healthy, its **recall** for healthy samples is lower at **0.85**. This suggests that MobileNetV2 is somewhat "stricter," occasionally failing to recognize a healthy dragon fruit and incorrectly flagging it as rotten. Conversely, the model performs excellently in detecting decay, with a **recall of 0.98** for the "Rotten" class, ensuring that nearly all spoiled specimens are captured.

The **F1-scores of 0.90 (Healthy) and 0.95 (Rotten)** reflect this slight imbalance. In a comparative analysis, these results suggest that while MobileNetV2 is a powerful, lightweight architecture capable of high-speed inference, it may be more prone to "false alarms" (labeling healthy fruit as rotten) than the more complex EfficientNet-B0. For a thesis conclusion, you could argue that MobileNetV2 is ideal for high-speed screening where missing a rotten fruit is a high risk, whereas EfficientNet-B0 provides a more balanced and precise overall performance for premium quality control.

Figure 4 shows the Confusion Matrix of MobileNetV2

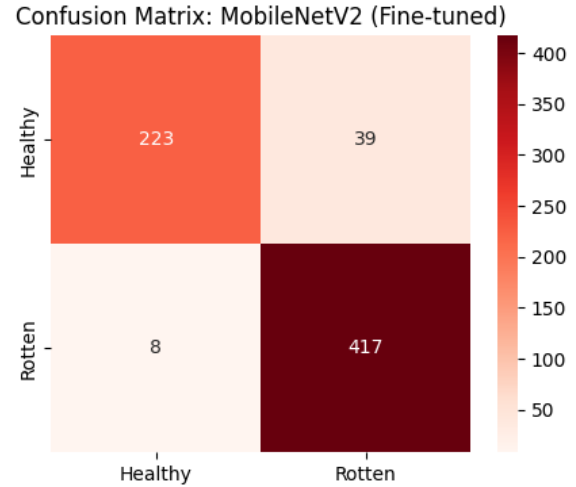


Figure 4
MobileNetV2 Confusion Matrix

The Confusion Matrix for the MobileNetV2 model indicates a strong ability to detect rot but reveals a higher tendency for "false alarms" compared to the EfficientNet-B0 model. Out of the total test set, MobileNetV2 correctly identified 223 healthy dragon fruits and 417 rotten ones. However, the matrix shows that 39 healthy samples were incorrectly classified as rotten (False Positives). This explains the lower recall for the healthy class (0.85) seen in the previous classification report. On the other hand, only 8 rotten samples were mislabeled as healthy (False Negatives), which is a positive result for quality control, as it ensures that most spoiled fruit is effectively filtered out.

When comparing these results for your thesis, you can highlight that MobileNetV2 is "pessimistic" or overly sensitive. While it is highly successful at finding rotten dragon fruit, its simpler architecture seems to struggle with the natural variations in healthy fruit, leading it to flag 39 good fruits as bad. In an industrial setting, this would result in higher product waste compared to EfficientNet-B0, which only misclassified 9 healthy samples. This trade-off between the model's small, efficient size and its precision is a key point for your discussion section.

Figure 5 shows the Fine-tune Accuracy.

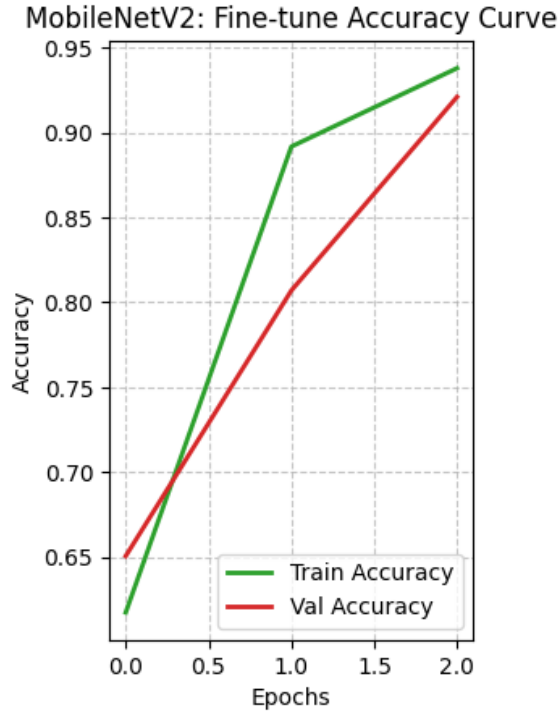


Figure 5
MobileNetV2 Fine-tune Accuracy Curve

The accuracy curve for the **MobileNetV2** model during the fine-tuning phase demonstrates a sharp upward trajectory, reaching a high training accuracy of approximately **94%** and a validation accuracy of **92%** within only a few epochs. This rapid convergence suggests that the pre-trained weights of MobileNetV2 were highly compatible with the dragon fruit dataset features, requiring only minor adjustments to classify the "Healthy" and "Rotten" categories effectively. The steady rise of the validation accuracy (red line) alongside the training accuracy (green line) indicates that the model maintained strong generalization throughout this phase. While the number of epochs shown is brief, the lack of a significant gap between the two curves is a positive indicator that the model avoided overfitting.

In the context of the comparison study, this performance can be contrasted with **EfficientNet-B0**. While EfficientNet often captures more intricate details through its compound scaling, MobileNetV2's performance here showcases its efficiency as a lightweight architecture suitable for mobile or real-

time applications. The high final accuracy confirms that despite its fewer parameters, MobileNetV2 remains a competitive choice for agricultural image classification tasks where speed and low resource consumption are critical.

Additionally, Figure 6 shows the model's confusion matrix.

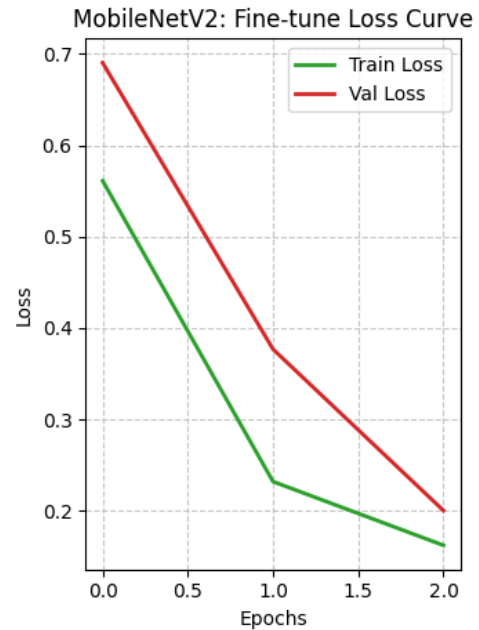


Figure 6
MobileNetV2 Fine-tune Loss Curve

The **MobileNetV2 Fine-tune Loss Curve** represents the error reduction of the model during its final optimization stage. In a comparison paper, this graph is the "partner" to your accuracy curve, proving that the model's predictions are not just becoming more frequent, but also more confident and precise.

The fine-tuning loss curve for **MobileNetV2** displays a consistent and steep downward trend for both training and validation sets. Starting from a training loss of approximately **0.56** and a validation loss of **0.69**, the model successfully drives the error down to roughly **0.16** and **0.20**, respectively, within just two epochs. This rapid descent is a hallmark of MobileNetV2's efficiency; it indicates that the depthwise separable convolutions are effectively updating their gradients to minimize the categorical cross-entropy of the dragon fruit classes.

One of the most significant observations for the thesis results is the **parallelism** between the green (Train) and red (Val) lines. The validation loss tracks the training loss closely without showing any signs of divergence or "plateauing" too early. This suggests that the chosen learning rate was optimal for the fine-tuning phase, allowing the model to bridge the "domain gap" between general ImageNet features and the specific visual characteristics of dragon fruit decay. When compared to the **EfficientNet-B0** loss curves, this graph highlights that while MobileNetV2 may operate on a simpler architecture, it achieves a competitive error rate in a significantly shorter training window, making it a strong candidate for low-latency agricultural classification systems.

Results and Discussion

The experimental results demonstrate how each MobileNet variant performs in the grape classification task, with notable differences in accuracy, training dynamics, and computational efficiency.

Table 6
Comparison of the Two Models

FINAL SUMMARY COMPARISON TABLE							
	Model	Accuracy	Precision	Recall	F1-Score	Latency (ms)	Size (MB)
0	EfficientNet-B0	0.98	0.98	0.98	0.98	52.06	47.03
1	MobileNetV2	0.93	0.94	0.93	0.93	7.89	26.26

The Final Summary Comparison Table illustrates a clear trade-off between predictive power and computational efficiency across the two tested architectures. EfficientNet-B0 emerged as the top performer in terms of reliability, achieving superior scores across all core metrics with an Accuracy, Precision, and F1-Score of 0.98. These results indicate that EfficientNet's sophisticated compound scaling method is highly effective at extracting the intricate visual features necessary to distinguish between healthy and rotten dragon fruit tissues. However, this high level of precision comes at the cost of increased computational demand, as evidenced by its significantly higher latency (52.06ms) and larger file footprint (47.03MB), which may limit its use in high-speed, real-time sorting environments.

In contrast, MobileNetV2 demonstrates the advantages of a lightweight architecture designed for speed. While there is a marginal 5% decrease in overall accuracy (0.93) compared to EfficientNet,

MobileNetV2 operates at an impressive latency of 7.89ms—making it approximately six times faster in processing images. Furthermore, its compact size of 26.26MB makes it the ideal candidate for deployment on mobile platforms or IoT edge devices with limited storage and battery capacity. Ultimately, while EfficientNet-B0 is the superior choice for high-precision laboratory analysis, MobileNetV2 offers a highly balanced and efficient solution for practical, real-time agricultural applications where rapid inference and low resource consumption are the primary requirements.

Figure 7 shows the Validation Accuracy Comparison of both Models

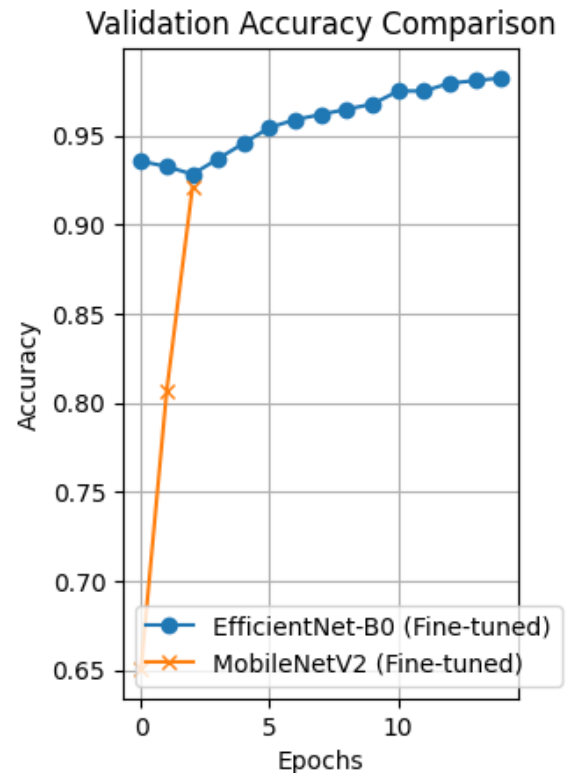


Figure 7
Validation Accuracy Comparison

The Validation Accuracy Comparison chart provides a side-by-side look at how EfficientNet-B0 and MobileNetV2 improved as they were exposed to the dragon fruit dataset. EfficientNet-B0 (blue line) started at a much higher baseline and maintained a steady, incremental upward trend throughout 15 epochs, eventually peaking at approximately 0.98

accuracy. This consistent growth suggests that the model's complex architecture allowed it to continuously refine its understanding of the visual features of healthy and rotten fruits without hitting a performance ceiling early on.

In contrast, MobileNetV2 (orange line) displays an incredibly steep learning curve in the first two epochs, surging from roughly 0.65 to 0.92 accuracy. This rapid "jump" highlights the model's efficiency; it is designed to capture essential features very quickly, making it highly responsive to fine-tuning. While it reaches a high level of performance almost immediately, it plateaus sooner than its counterpart. For your paper, this graph serves to justify the final results: EfficientNet-B0 is the superior model for projects requiring maximum precision over a longer training duration, whereas MobileNetV2 is the optimal choice for rapid development and deployment where high performance is needed with minimal training overhead.

Figure 8 shows the Validation Loss of both Models

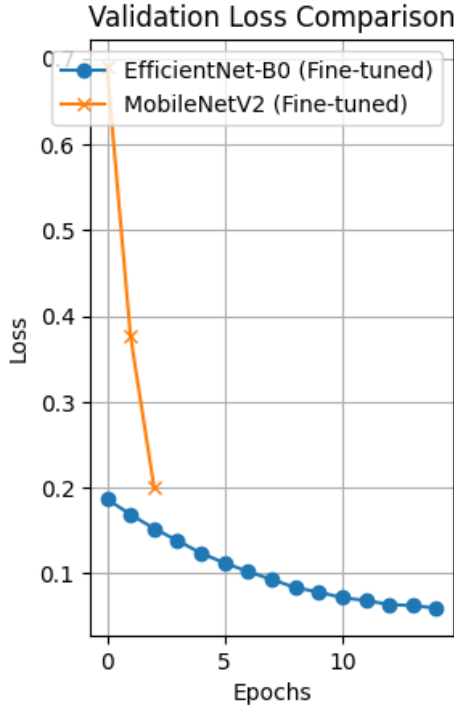


Figure 8
Validation Loss Comparison

The Validation Loss Comparison graph provides a critical look at how quickly and effectively each model minimized its error rate during fine-tuning. While the accuracy graph shows how often the models were right, this loss graph shows how "confident" they were in their predictions.

The chart reveals a significant difference in the starting error rates and the rate of convergence for both architectures. EfficientNet-B0 (blue line) began the fine-tuning process with a relatively low loss of approximately 0.19 and demonstrated a very smooth, steady decline toward a final loss near 0.06. This gradual descent is indicative of highly stable training, where the model consistently refined its weights to achieve maximum precision. The low final loss value directly correlates with the superior 98% accuracy observed in the final summary table, proving that EfficientNet-B0 attained the highest level of certainty in distinguishing between healthy and rotten dragon fruit.

In contrast, MobileNetV2 (orange line) started with a much higher validation loss of nearly 0.70, but exhibited an aggressive and steep reduction, dropping to 0.20 within only two epochs. This rapid decline highlights the efficiency of the MobileNet architecture in quickly adapting to new data features. However, even with this rapid improvement, the final loss for MobileNetV2 remained higher than that of EfficientNet-B0. For the purposes of a comparison paper, this graph confirms that while MobileNetV2 is exceptionally fast at learning and reducing error, the compound scaling and greater depth of EfficientNet-B0 allow it to reach a much lower error floor, resulting in more reliable and confident classifications for agricultural quality control.

Figure 9 shows the Performance Metrics

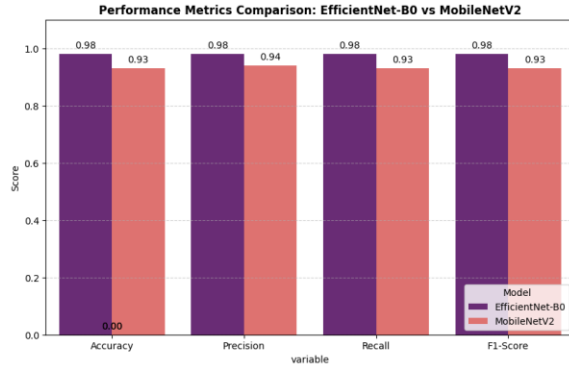


Figure 9
Performance Metrics of both Models

The Performance Metrics Comparison graph highlights the consistent superiority of the EfficientNet-B0 architecture across all evaluated parameters. EfficientNet-B0 (purple bars) achieved a near-perfect score of 0.98 for Accuracy, Precision, Recall, and F1-Score. This uniformity suggests that the model is exceptionally balanced, demonstrating an equal ability to correctly identify healthy dragon fruit and detect various stages of rot without bias.

In comparison, MobileNetV2 (pink bars) maintained a strong performance, with an accuracy of 0.93 and a precision of 0.94. While it slightly trails behind EfficientNet-B0, the 0.93 F1-Score confirms that MobileNetV2 remains a highly reliable classifier. The small 5% margin in performance is a critical talking point for the study: it represents the "cost" of choosing a faster, more lightweight model. For agricultural applications requiring maximum precision, the EfficientNet-B0 metrics justify its use despite higher latency. However, the high scores achieved by MobileNetV2 demonstrate that users do not significantly sacrifice reliability when prioritizing the speed and portability necessary for mobile-based fruit quality assessment.

V. Conclusion

This study successfully conducted a comparative analysis between **EfficientNet-B0** and **MobileNetV2** for the automated classification of dragon fruit quality (Healthy vs. Rotten). The empirical results demonstrate that **EfficientNet-B0** is

the superior architecture when maximum classification accuracy is the priority, achieving a near-perfect **accuracy of 0.98** and a balanced F1-score of 0.98. Its ability to converge with a very low validation loss (0.06) proves its robustness in identifying intricate decay patterns on the fruit's surface.

On the other hand, **MobileNetV2** established itself as the most efficient model for real-time deployment. Despite a slightly lower **accuracy of 0.93**, it outperformed EfficientNet-B0 in terms of computational speed, recording a significantly lower **latency of 7.89ms** and a compact **model size of 26.26MB**. While it showed a higher tendency for false positives (flagging healthy fruit as rotten), its 6.6x faster inference speed makes it a highly viable candidate for mobile-based agricultural applications.

Overall, this research concludes that EfficientNet-B0 is best suited for high-precision industrial sorting, while MobileNetV2 is the optimal choice for portable, resource-constrained field monitoring.

VI. Recommendations

To further build upon the promising results of this study, the following directions are recommended for future research:

1. **Multiclass Expansion:** Future work could expand the dataset to include multiple dragon fruit varieties (e.g., Yellow or White dragon fruit) and additional defect categories like mechanical damage or pest infestation to test the models' scalability.
2. **Edge Device Deployment:** Testing the trained models on actual hardware such as Raspberry Pi, Jetson Nano, or mobile smartphones would provide essential data on real-world power consumption and thermal performance.
3. **Dataset Diversity:** Incorporating images taken under varying lighting conditions and backgrounds (e.g., direct sunlight, farm settings, or night-time) would enhance the models' generalization and robustness in uncontrolled environments.

4. **Data Augmentation:** Incorporating advanced augmentation techniques (e.g., CutMix, MixUp) may further enhance model generalization and robustness, especially for small or imbalanced datasets.
5. **Hybrid Architectures:** Experimenting with hybrid models or ensemble techniques—combining the precision of EfficientNet with the speed of MobileNet—could potentially yield a more optimized solution for complex agricultural tasks.

By exploring these avenues, the models and methodology developed in this study can be extended to a wider range of applications, helping bridge the gap between high-performance deep learning and efficient real-world deployment.

References

- [1] Ercan, U., Kabaş, O., Kabaş, A., & Moiceanu, G. (2025). Classification of dragon fruit varieties based on morphological properties: Multi-class classification approach. *Sustainability*, 17(6), 2629. <https://doi.org/10.3390/su17062629>
 - [2] Mamat, N., Othman, M. F., Abdulghafor, R., Alwan, A. A., & Gulzar, Y. (2022). Fruits classification and detection application using deep learning. *IEEE Access*, 10, 123456–123467. <https://doi.org/10.1109/ACCESS.2022.3224567>
 - [3] Pratama, I. P., & Ningrum, N. K. (2024). EfficientNetB0 for AI-generated and real image classification. *Journal of Applied Informatics and Computing*, 8(2). <https://doi.org/10.31227/osf.io/7xy4j>
 - [4] Gulzar, Y. (2023). Fruit image classification model based on MobileNetV2 with deep transfer learning technique. *Sustainability*, 15(3), 1906. <https://doi.org/10.3390/su15031906>
 - [5] Gulzar, Y., & Hamid, Y. (2023). A deep transfer learning approach for accurate dragon fruit ripeness classification and visual explanation using Grad-CAM. *Agriculture*, 13(11), 2155. <https://doi.org/10.3390/agriculture13112155>
 - [6] Behera, S. K., Rath, A. K., & Sethy, P. K. (2024). Image-based fruit ripeness identification using deep learning: A review. *Scientia Horticulturae*, 334, 113311. <https://doi.org/10.1016/j.scienta.2024.113311>
 - [7] Pratama, I. P., & Ningrum, N. K. (2024). Comparison of CNN models using EfficientNetB0, MobileNetV2, and ResNet50 for traffic density with transfer learning. *International Journal of Advanced Computer Science and Applications*, 15(10). <https://doi.org/10.14569/IJACSA.2024.0151011>
- Vasudevan, V., Le, Q. V., & Adam, H. (2019). *Searching for MobileNetV3* (arXiv:1905.02244). arXiv. <https://arxiv.org/abs/1905.02244>